SSC MTS 2024 question paper with answers

28 October 2024 · Shift 3

79 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 11 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 16 questions

Q.1

The LCM of 3,4 and 5 is:

(A) 30
(B) 20
(C) 60
(D) 48

Q.3

A motor car starts with a speed of 60 km/h and increases its speed after every two hours by 15 km/h. In how much time will it cover a distance of 450 km?

(A) 5 hours
(B) 6 hours
(C) 2 hours
(D) 9 hours

Q.4

In an election between two candidates, one person got 35% of the votes and lost by 6 votes. Assuming that there were no invalid votes, what was the total number of votes?

(A) 20
(B) 30
(C) 18
(D) 50

Q.5

By selling an article for ₹19.5, a dealer makes a profit of 30%. By how much should he increase his selling price (in ₹) so as to make a profit of 35%?

(A) 2.75
(B) 0.75
(C) 1.75
(D) 3.75

Q.7

A chair was sold for ₹8,601 after giving two successive discounts of 8.5% and 6%. Find the marked price.

(A) ₹9,612
(B) ₹9,000
(C) ₹9,848
(D) ₹10,000

Q.8

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 4% per year rate of interest for 3 years.

(A) 360
(B) 380
(C) 420
(D) 460

Q.9

In a circular race of 1100 m, P and Q start from the same point, at the same time and at speeds of 36 km/h and 63 km/h, respectively. When will they meet again for the first time on the track if they are running in the opposite direction?

(A) 27 s
(B) 20 s
(C) 10 s
(D) 40 s

Q.11

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 7% per year rate of interest for 5 years.

(A) 1185
(B) 1225
(C) 1165
(D) 1265

Q.13

Sunita borrows ₹5,500 for 3 years at 6.5% per annum simple interest. She immediately lends it to Vedanshi at 9.3% per annum simple interest for 3 years. Find her gain (in ₹) in the transaction per year.

(A) 161
(B) 149
(C) 157
(D) 154

Q.14

If the total surface area of a hemisphere is 192π cm^2, then the radius of the base of the hemisphere is:

(A) 64 cm
(B) 4 cm
(C) 8 cm
(D) 16 cm

Q.15

How much water should be added to a 30 l mixture of milk and water, which currently has a ratio of 7 : 3, so that the resultant mixture contains 40% water?

(A) 6 l
(B) 4.25 l
(C) 3 l
(D) 5 l

Q.16

In a ration shop, there is a certain amount of cereals to distribute to 42 families for 15 days. If 48 kg of cereals can be distributed to 30 families for 7 days, then the quantity of cereals already in the shop was:

(A) 84 kg
(B) 96 kg
(C) 112 kg
(D) 144 kg

Q.17

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 72 kg and 35 kg is:

(A) 50.4
(B) 52.4
(C) 49.4
(D) 51.4

Q.18

The mean of five numbers is 55. If one of the numbers is excluded, the mean gets reduced by 5. Which of the following is the excluded number?

(A) 75
(B) 55
(C) 65
(D) 50

Q.19

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 68 kg and 39 kg is:

(A) 49.4
(B) 51.4
(C) 50.4
(D) 52.4

Q.20

A cylindrical glass with a diameter of 20 cm is having the juice to a height of 9 cm. An ice cube of size 8 cm is amerced in it completely. How much will the juice level rise in the glass? (Use π = 3.142 and answer correctly up to the nearest mm)

(A) 17 mm
(B) 163 mm
(C) 16 mm
(D) 63 mm

Reasoning Ability and Problem Solving 15 questions

Q.1

In a code language, 'PRETTY' is written as 'BIVGGK' and 'JOYFUL' is written as 'OLBUFQ'. How will 'EXOTIC' be written in the same language?

(A) XCLGRV
(B) SBLGRV
(C) SCLGRV
(D) XBLGRV

Q.2

EJIA is related to FKJB in a certain way based on the English alphabetical order. In the same way, ZDCU is related to AEDV. To which of the following is OTSK related, following the same logic?

(A) LTPU
(B) PULT
(C) PTLU
(D) PUTL

Q.4

What will come in the place of '?' in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 110 + 54 × 9 – 13 ÷ 2 = ?

(A) 144
(B) 114
(C) 130
(D) 136

Q.5

Select the number from among the given options that can replace the question mark (?) in the following series. 9, 45, 225, 1125, 5625, ?

(A) 4679
(B) 7689
(C) 28125
(D) 2799

Q.7

Seven friends - Asha, Sreeja, Mamata, Ramya, Hema, Jaya, and Mohini - are sitting around a circular table, facing the centre (but not necessarily in the same order). Ramya is sitting fourth to the right of Asha. Mamata sits third to the left of Ramya. Jaya has Sreeja and Mamata as immediate neighbours. Hema is sitting to the immediate right of Ramya. Who is sitting second to the left of Mohini?

(A) Asha
(B) Ramya
(C) Hema
(D) Sreeja

Q.8

MIPL is related to OKRN in a certain way based on the English alphabetical order. In the same way, PLSO is related to RNUQ. To which of the following is HDKG related, following the same logic?

(A) JFMI
(B) FJMI
(C) FJIM
(D) JFIM

Q.9

What will come in place of the question mark (?) in the following equation if '+' and '–' are interchanged and '×' and '÷' are interchanged? 62 – 36 + 18 ÷ 6 × 3 = ?

(A) 107
(B) 62
(C) 27
(D) 17

Q.10

What should come in place of the question mark (?) in the following series based on the English alphabetical order? RJX, PNV, NRT, LVR ,?

(A) KZP
(B) JZQ
(C) JZP
(D) KAP

Q.11

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All Excels are Words. All Words are PowerPoints. Conclusions: (I) All PowerPoints are Excels. (II) At least some Words are Excels.

(A) Only conclusion I follows
(B) Neither conclusion I nor II follows
(C) Only conclusion II follows
(D) Both conclusion I and II follow

Q.14

D, E, F, L, R, S and T are sitting around a circular table facing the centre(but not necessarily in the same order). R sits second to the right of D. L sits to the immediate left of D. T sits third to the left of L. F sits third to the left of S. Neither E nor S is an immediate neighbour of R. What is the position of S with respect to R?

(A) Third to the right
(B) Second to the right
(C) Fifth to the right
(D) Fourth to the left

Q.15

In a certain code language, 'LINK' is coded as '4175' and 'NICK' is coded as '7419'. What is the code for 'C' in the given language?

(A) 1
(B) 7
(C) 9
(D) 4

Q.16

35 is related to 105 following a certain logic. Following the same logic, 51 is related to 153. To which of the following is 72 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying, etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 214
(B) 216
(C) 212
(D) 210

Q.17

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statement(s). Statements: All bats are mammals. No mammal is a fish. Conclusions: (I) No bat is a fish. (II) Some fishes are mammals.

(A) Only conclusion (I) follows.
(B) Neither conclusion (I) nor (II) follows.
(C) Both conclusions (I) and (II) follow.
(D) Only conclusion (II) follows.

Q.19

In a certain code language, 'VOLT' is coded as '5731' and 'LEAD' is coded as '4186'. What is the code for 'L' in the given code language?

(A) 1
(B) 3
(C) 8
(D) 4

Q.20

In the following number-pairs, the second number is obtained by applying certain mathematical operations to the first number. Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 25, 77; 28, 86

(A) 34, 103
(B) 34, 104
(C) 35, 104
(D) 35, 103

General Awareness 23 questions

Q.1

When was the first-round table Conference started by the British Government in London?

(A) November 1930
(B) December 1931
(C) December 1929
(D) November 1932

Q.2

Who is literate according to the Census of India?

(A) A person above 5 years of age, who can read and write with understanding in any language.
(B) A person above 7 years of age, who can read and write with understanding in any language.
(C) A person above 18 years of age, who can read and write with understanding in any language.
(D) A person above 14 years of age, who can read and write with understanding in any language.

Q.3

Rank the following states in ascending order, based on their population density, as per Census 2011. Tripura, Meghalaya, Manipur, Nagaland

(A) Nagaland-Manipur-Meghalaya-Tripura
(B) Nagaland-Tripura-Manipur-Meghalaya
(C) Meghalaya-Tripura-Manipur-Nagaland
(D) Meghalaya-Tripura-Nagaland-Manipur

Q.4

Who does NOT fall under the executive of the Indian Union?

(A) Chief Justice of India
(B) Council of Ministers
(C) Prime Minister
(D) President

Q.5

'Rajatarangini', a vivid account of the kings of Kashmir, was composed by Kalhana in the _____ language.

(A) Urdu
(B) Sanskrit
(C) Arabic
(D) Persian

Q.6

'The Climate Smart Cities Assessment Framework 2.0' is to evaluate how many cities across India?

(A) 100
(B) 32
(C) 70
(D) 120

Q.7
The East India Company lost all its administrative powers with the enforcement of _____.

(A) The Indian Councils Act, 1892
(B) The Government of India Act, 1858
(C) The Charter Act of 1853
(D) The Regulating Act, 1773

Q.8
In which state of India is the Telaiya dam located?

(A) Jharkhand
(B) Karnataka
(C) Madhya Pradesh
(D) Bihar

Q.9
When a river reaches a low lying plain, it continues erosion of soil as well as deposition and as a result of both these factors another river channel develops and later it is detached from the main river forming a U-shaped lake; this is called:

(A) an oxbow lake
(B) a gulf
(C) a canal
(D) an estuary

Q.10
Which of the following awards and recognitions was NOT received by Vempati Chinna Satyam?

(A) Kalidas Samman
(B) Padma Shri
(C) Honorary doctorate
(D) Sangeet Natak Akadami Award

Q.11
As of February 2023, who is the Union Minister of Road Transport and Highways?

(A) Dharmendra Pradhan
(B) Nitin Gadkari
(C) Ashwini Vaishnav
(D) Mansukh Mandaviya

Q.12
The autobiography of which of the following famous Indian musicians is titled ‘My Life, My Music’?

(A) Zakir Hussain
(B) Bismillah Khan
(C) Ravi Shankar
(D) Shivkumar Sharma

Q.14
Which of the following dances is performed in Goa during the festival Ganesh Chaturthi?

(A) Dhangar
(B) Dhalo
(C) Tarangmel
(D) Fugdi

Q.15
Sultan Ghari tomb is situated to west of the Qutub in Delhi is like a fortress within a walled enclosure with bastions on the comers, with its _____ grave chamber underground.

(A) rectangular
(B) pentagonal
(C) octagonal
(D) squarish

Q.16
The official journey of 'Azadi Ka Amrit Mahotsav' commenced on:

(A) 12th March 2021
(B) 15th August 2022
(C) 14th August 2022
(D) 14th April 2021

Q.17
Which of the following is the role of fibre in the diet?

(A) Promoting digestion and bowel regularity
(B) Regulating body temperature
(C) Enhancing taste of food
(D) Providing quick energy

Q.18

Which of the following is NOT a feature of the contingency fund of India?

(A) It is established under Article 267.

(B) It is for emergencies.

(C) Allowances and emoluments of the President are charged under it.

(D) It is issued on behalf of the President of India.

Q.19

Which of the following is a unitary feature of the Indian Constitution?

(A) single citizenship

(B) Division of Powers

(C) Bicameralism

(D) Written Constitution

Q.21

A three-time Olympian, PT Usha broke the Asian Games record in four events (200 m, 400 m, 400 m hurdles and 4 × 400 m relay) in _____.

(A) Beijing, 1990

(B) Seoul, 1986

(C) New Delhi, 1982

(D) Hiroshima, 1994

Q.22

Fa-Hsien, a Chinese Buddhist Pilgrim, began his journey back home (China) from _____.

(A) Bengal

(B) Taxila

(C) Prayag

(D) Arikamedu

Q.23

In which of the following states can the 'Loo' phenomenon be observed?

(A) Rajasthan

(B) Goa

(C) Tamil Nadu

(D) Manipur

Q.24

At which Olympic games did India receive its first Olympic medal in a shooting event?

(A) London 2012

(B) Beijing 2008

(C) Tokyo 2022

(D) Athens 2004

Q.25

Lalita J Rao is associated with which of the following gharanas?

(A) Agra gharana

(B) Kirana gharana

(C) Benaras gharana

(D) Jaipur gharana

English Language and Comprehension 25 questions

Q.1

Select the most appropriate ANTONYM of the given word. Frugal

(A) Profligate

(B) Thrifty

(C) Cheap

(D) Happy

Q.2

Select the sentence that has a grammatical error.

(A) It was a huge zoo with an elephant enclosure.

(B) Suhas has a younger brother and an elder sister.

(C) He was an honest engineer by profession.

(D) Alka bought new pair of shoes for the party.

Q.3
Select the INCORRECTLY spelt word.

(A) Admited
(B) Admittance
(C) Admissible
(D) Admission

Q.4
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The athletes / trained tirelessness / to prepare for / the upcoming competition.

(A) The athletes
(B) trained tirelessness
(C) the upcoming competition
(D) to prepare for

Q.5
Select the most appropriate option to fill in the blank. The storm caused extensive damage to homes, _____ many families homeless.

(A) is left
(B) to leave
(C) leave
(D) leaving

Q.6
Select the correct spelling of the incorrectly spelt word in the given sentence. He was suspended from the party due to his suspiscious activities.

(A) Suspicius
(B) Suspeceous
(C) Suspitious
(D) Suspicious

Q.7
Select the most appropriate option that can substitute the underlined segment in the given sentence. The elusive moon peeked from behind the clouds, “illuminating the earth with” its gentle glow.

(A) empowering the earth via
(B) glowing up the earth for
(C) encouraging the earth with
(D) lighting up the earth with

Q.8
Select the most appropriate meaning of the given idiom. Grease someone's palm

(A) To flatter someone
(B) To abuse someone
(C) To bribe someone
(D) To beat someone

Q.9
Select the most appropriate synonym of the given word. Hermetic

(A) Obnoxious
(B) Lucid
(C) Intelligible
(D) Esoteric

Q.10
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. She is learning / to play the guitar / to pursue her passion / for the music.

(A) to play the guitar
(B) to pursue her passion
(C) for the music.
(D) She is learning

Q.11
Select the most appropriate ANTONYM of the given word. Ambrosial

(A) Pleasing
(B) Faithful
(C) Malodorous
(D) Vengeful

Q.12
The following sentence has been split into four segments. Identify the segment that contains a grammatical/spelling error. He was lying at / his bed when he / saw the snake entering / through the window.

(A) his bed when he
(B) He way lying at
(C) saw the snake entering
(D) through the window

Q.13
Select the most appropriate ANTONYM of the underlined word. she had “garnered” the information from various books.

(A) Liberated
(B) Criticised
(C) Dispersed
(D) Implemented

Q.14
Select the correct idiom to substitute the underlined segment in the following sentence. His remarks “made the matter worse.”

(A) Made a splash
(B) Added fuel to the fire
(C) Killed two birds with one stone
(D) Spilled the beans

Q.15
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. He is / an expert / in / an English literature.

(A) an expert
(B) He is
(C) in
(D) an English literature

Q.16
Select the option that rectifies the INCORRECT spelling in the given sentence. Pleasee allow me to start.

(A) Pleasse
(B) Pleeasse
(C) Please
(D) Pleesse

Q.17
Select the most appropriate option that can substitute the underlined segment in the given sentence. Do you know the purpose of a fire extinguisher? It helps to “put up with” an engulfing fire, in case of an emergency.

(A) turn over
(B) stand out
(C) run away with
(D) put out

Q.18
Select the most appropriate meaning of the underlined phrase. Those who have died cannot be “brought back to life”.

(A) Abstract
(B) Retract
(C) Extract
(D) Resurrect

Q.19
Select the most appropriate option that can substitute the underlined segment in the given sentence. A raven, which you know is black as coal, was envious of the swan, because her “feathers are as white as” the purest snow.

(A) feathers is a white as
(B) feathers was a white as
(C) feathers were as white as
(D) feathers is as white as

Q.20
Select the most appropriate synonym of the given word. Vivid

(A) Soothing
(B) Varied
(C) Bright
(D) Dull

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Khashaba Jadhav was the independent India's first individual olympic medallist. There (1) _____ some firsts that stay in history forever. The one held by Khashaba Jadhav is one such first. He might have been referred to as the Pocket Dynamo, but his deeds dwarfed those of far (2) _____ athletes in his sport. Just a couple of years after India had become a Republic, Jadhav took part in the 1952 Olympics at Helsinki, Finland. As funding sportspersons weren't really a (3) _____ for a country just finding its feet, Jadhav's trip to Helsinki was actually funded by the public. Jadhav however, (4) _____ his way to fame, becoming independent India's first individual Olympic medal winner. He (5) _____ bronze in the bantamweight category.

Select the most appropriate option to fill in blank 1.

(A) are
(B) is
(C) were
(D) was

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Khashaba Jadhav was the independent India's first individual olympic medallist. There (1) _____ some firsts that stay in history forever. The one held by Khashaba Jadhav is one such first. He might have been referred to as the Pocket Dynamo, but his deeds dwarfed those of far (2) _____ athletes in his sport. Just a couple of years after India had become a Republic, Jadhav took part in the 1952 Olympics at Helsinki, Finland. As funding sportspersons weren't really a (3) _____ for a country just finding its feet, Jadhav's trip to Helsinki was actually funded by the public. Jadhav however, (4) _____ his way to fame, becoming independent India's first individual Olympic medal winner. He (5) _____ bronze in the bantamweight category.

Select the most appropriate option to fill in blank 2.

(A) bigger
(B) big
(C) small
(D) smaller

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Khashaba Jadhav was the independent India's first individual olympic medallist. There (1) _____ some firsts that stay in history forever. The one held by Khashaba Jadhav is one such first. He might have been referred to as the Pocket Dynamo, but his deeds dwarfed those of far (2) _____ athletes in his sport. Just a couple of years after India had become a Republic, Jadhav took part in the 1952 Olympics at Helsinki, Finland. As funding sportspersons weren't really a (3) _____ for a country just finding its feet, Jadhav's trip to Helsinki was actually funded by the public. Jadhav however, (4) _____ his way to fame, becoming independent India's first individual Olympic medal winner. He (5) _____ bronze in the bantamweight category.

Select the most appropriate option to fill in blank 3.

(A) supremacy
(B) subsequence
(C) posteriority
(D) priority

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Khashaba Jadhav was the independent India's first individual olympic medallist. There (1) _____ some firsts that stay in history forever. The one held by Khashaba Jadhav is one such first. He might have been referred to as the Pocket Dynamo, but his deeds dwarfed those of far (2) _____ athletes in his sport. Just a couple of years after India had become a Republic, Jadhav took part in the 1952 Olympics at Helsinki, Finland. As funding sportspersons weren't really a (3) _____ for a country just finding its feet, Jadhav's trip to Helsinki was actually funded by the public. Jadhav however, (4) _____ his way to fame, becoming independent India's first individual Olympic medal winner. He (5) _____ bronze in the bantamweight category.

Select the most appropriate option to fill in blank 4.

(A) buckled-under

(B) Surrendered

(C) wrestled

(D) Lazed

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Khashaba Jadhav was the independent India's first individual olympic medallist. There (1) _____ some firsts that stay in history forever. The one held by Khashaba Jadhav is one such first. He might have been referred to as the Pocket Dynamo, but his deeds dwarfed those of far (2) _____ athletes in his sport. Just a couple of years after India had become a Republic, Jadhav took part in the 1952 Olympics at Helsinki, Finland. As funding sportspersons weren't really a (3) _____ for a country just finding its feet, Jadhav's trip to Helsinki was actually funded by the public. Jadhav however, (4) _____ his way to fame, becoming independent India's first individual Olympic medal winner. He (5) _____ bronze in the bantamweight category.

Select the most appropriate option to fill in blank 5.

(A) constructed

(B) lulled

(C) forgave

(D) claimed

Answer key

Numerical and Mathematical Ability

Q.1 **(C)**	Q.3 **(B)**	Q.4 **(A)**	Q.5 **(B)**	Q.7 **(D)**	Q.8 **(C)**
Q.9 **(D)**	Q.11 **(B)**	Q.13 **(D)**	Q.14 **(C)**	Q.15 **(D)**	Q.16 **(D)**
Q.17 **(A)**	Q.18 **(A)**	Q.19 **(C)**	Q.20 **(C)**		

Reasoning Ability and Problem Solving

Q.1 **(A)**	Q.2 **(D)**	Q.4 **(C)**	Q.5 **(C)**	Q.7 **(B)**	Q.8 **(A)**
Q.9 **(B)**	Q.10 **(C)**	Q.11 **(C)**	Q.14 **(B)**	Q.15 **(C)**	Q.16 **(B)**
Q.17 **(A)**	Q.19 **(A)**	Q.20 **(B)**			

General Awareness

Q.1 **(A)**	Q.2 **(B)**	Q.3 **(A)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(A)**
Q.7 **(B)**	Q.8 **(A)**	Q.9 **(A)**	Q.10 **(B)**	Q.11 **(B)**	Q.12 **(C)**
Q.14 **(D)**	Q.15 **(C)**	Q.16 **(A)**	Q.17 **(A)**	Q.18 **(C)**	Q.19 **(A)**
Q.21 **(B)**	Q.22 **(A)**	Q.23 **(A)**	Q.24 **(D)**	Q.25 **(A)**	

English Language and Comprehension

Q.1 **(A)**	Q.2 **(D)**	Q.3 **(A)**	Q.4 **(B)**	Q.5 **(D)**	Q.6 **(D)**
Q.7 **(D)**	Q.8 **(C)**	Q.9 **(D)**	Q.10 **(C)**	Q.11 **(C)**	Q.12 **(B)**
Q.13 **(C)**	Q.14 **(B)**	Q.15 **(D)**	Q.16 **(C)**	Q.17 **(D)**	Q.18 **(D)**
Q.19 **(C)**	Q.20 **(C)**	Q.21 **(A)**	Q.22 **(A)**	Q.23 **(D)**	Q.24 **(C)**
Q.25 **(D)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (C) 60

3, 4 and 5 have no common factor, so their LCM is their product: 3 × 4 × 5 = 60.

Q.3 · Answer (B) 6 hours

In the first two hours the car covers 2 × 60 = 120 km, in the next two 2 × 75 = 150 km and in the two after that 2 × 90 = 180 km: 450 km in 6 hours.

Q.4 · Answer (A) 20

The loser got 35% and the winner 65%, so the margin of 30% of the votes is 6 votes and the total is 6/0.3 = 20.

Q.5 · Answer (B) 0.75

Cost price = 19.5/1.3 = ₹15. At 35% profit the price is 15 × 1.35 = ₹20.25, an increase of ₹0.75.

Q.7 · Answer (D) ₹10,000

The two discounts leave 0.915 × 0.94 = 0.8601 of the marked price, so the marked price is 8,601/0.8601 = ₹10,000.

Q.8 · Answer (C) 420

SI = 3,500 × 4 × 3/100 = ₹420.

Q.9 · Answer (D) 40 s

36 km/h = 10 m/s and 63 km/h = 17.5 m/s. Running in opposite directions they close the 1,100 m track at 27.5 m/s, so they first meet after 1,100/27.5 = 40 s.

Q.11 · Answer (B) 1225

SI = 3,500 × 7 × 5/100 = ₹1,225.

Q.13 · Answer (D) 154

She pays 6.5% and earns 9.3% on ₹5,500, a gain of 2.8% a year: 5,500 × 2.8/100 = ₹154.

Q.14 · Answer (C) 8 cm

A hemisphere's total surface area is $2\pi r^2 + \pi r^2 = 3\pi r^2 = 192\pi$, so $r^2 = 64$ and r = 8 cm.

Q.15 · Answer (D) 5 l

The 30 l holds 21 l of milk and 9 l of water. For water to be 40%, the 21 l of milk must be 60% of the mixture, which is then 35 l: add 35 − 30 = 5 l of water.

Q.16 · Answer (D) 144 kg

48 kg feeds 30 × 7 = 210 family-days, so 42 × 15 = 630 family-days need 48 × 630/210 = 144 kg.

Q.17 · Answer (A) 50.4

(40 + 49 + 56 + 72 + 35)/5 = 252/5 = 50.4 kg.

Q.18 · Answer (A) 75

The five numbers add up to 5 × 55 = 275 and the remaining four to 4 × 50 = 200, so the excluded number is 75.

Q.19 · Answer (C) 50.4

(40 + 49 + 56 + 68 + 39)/5 = 252/5 = 50.4 kg.

Q.20 · Answer (C) 16 mm

The cube's volume, $8^3 = 512$ cm³, spreads over the glass's base of $\pi \times 10^2 = 314.2$ cm², raising the juice by 512/314.2 = 1.63 cm: about 16 mm.

Reasoning Ability and Problem Solving

Q.1 · Answer (A) XCLGRV

The first and last letters swap, then each letter is replaced by its opposite in the alphabet (A ↔ Z, B ↔ Y, ...): PRETTY → YRETTP → BIVGGK and JOYFUL → LOYFUJ → OLBUFQ. So EXOTIC → CXOTIE → XCLGRV.

Q.2 · Answer (D) PUTL

Each letter moves 1 place forward (Z wraps round to A): EJIA → FKJB and ZDCU → AEDV. So OTSK → PUTL.

Q.4 · Answer (C) 130

After interchanging the signs: 110 – 54 ÷ 9 + 13 × 2 = 110 – 6 + 26 = 130.

Q.5 · Answer (C) 28125

Each term is 5 times the one before: 5,625 × 5 = 28,125.

Q.7 · Answer (B) Ramya

Going round to the left from Asha the order is Asha, Mohini, Hema, Ramya, Sreeja, Jaya, Mamata: Ramya is fourth to Asha's right, Mamata third to Ramya's left, Hema to Ramya's immediate right, and Jaya sits between Sreeja and Mamata. Second to the left of Mohini is Ramya.

Q.8 · Answer (A) JFMI

Each letter moves 2 places forward: MIPL → OKRN and PLSO → RNUQ. So HDKG → JFMI.

Q.9 · Answer (B) 62

After interchanging the signs: 62 + 36 – 18 × 6 ÷ 3 = 62 + 36 – 36 = 62.

Q.10 · Answer (C) JZP

First letters move back 2 (R, P, N, L, so J), second letters forward 4 (J, N, R, V, so Z) and third letters back 2 (X, V, T, R, so P): JZP.

Q.11 · Answer (C) Only conclusion II follows

'All Excels are Words' converts to 'some Words are Excels', so II follows. All Excels being PowerPoints does not make every PowerPoint an Excel, so I does not follow.

Q.14 · Answer (B) Second to the right

Going round to the left from D the order is D, L, E, S, T, R, F. R is second to D's right, L to D's immediate left and T third to L's left; F, third to the left of S, must take the seat next to R, since neither E nor S may, which puts S between E and T. S is second to the right of R.

Q.15 · Answer (C) 9

I, N and K are in both words, as are 4, 7 and 1 in both codes. The letter left in NICK is C and the digit left in its code is 9, so C = 9.

Q.16 · Answer (B) 216

Each number is related to 3 times itself: 35 × 3 = 105 and 51 × 3 = 153. Likewise 72 × 3 = 216.

Q.17 · Answer (A) Only conclusion (I) follows.

All bats are mammals and no mammal is a fish, so no bat is a fish (I). 'No mammal is a fish' rules out II.

Q.19 · Answer (A) 1

L is the only letter in both words and 1 the only digit in both codes, so L = 1.

Q.20 · Answer (B) 34, 104

The second number is 3 times the first plus 2: 25 × 3 + 2 = 77 and 28 × 3 + 2 = 86. Likewise 34 × 3 + 2 = 104.

General Awareness

Q.1 · Answer (A) November 1930

The First Round Table Conference opened in London on 12 November 1930 and ran until January 1931; the Congress stayed away.

Q.2 · Answer (B) A person above 7 years of age, who can read and write with understanding in any language.
The Census counts a person aged seven or above who can both read and write with understanding in any language as literate.

Q.3 · Answer (A) Nagaland-Manipur-Meghalaya-Tripura
SSC's key follows the Census 2011 densities (persons per km²) of Nagaland 119, Manipur 128, Meghalaya 132 and Tripura 350. (Some tables give Manipur 115, leaving out three subdivisions of Senapati district, which would put it first, an order no option offers.)

Q.4 · Answer (A) Chief Justice of India
The Union executive includes the President and the Council of Ministers headed by the Prime Minister. The Chief Justice of India heads the judiciary.

Q.5 · Answer (B) Sanskrit
Kalhana wrote the 'Rajatarangini' in Sanskrit verse in the 12th century.

Q.6 · Answer (A) 100
SSC's key gives 100: the framework was set up under the Smart Cities Mission to benchmark the climate actions of its 100 Smart Cities, and the Ministry of Housing and Urban Affairs launched version 2.0 in September 2020. (Its 2020 assessment in fact covered 126 cities, the 100 Smart Cities among them.)

Q.7 · Answer (B) The Government of India Act, 1858
The Government of India Act, 1858, passed after the revolt of 1857, ended the Company's rule and transferred the government of India to the Crown.

Q.8 · Answer (A) Jharkhand
The Tilaiya (Telaiya) dam, the first dam of the Damodar Valley Corporation, is on the Barakar river in Koderma district, Jharkhand.

Q.9 · Answer (A) an oxbow lake
A meander cut off from the main channel by erosion and deposition forms an oxbow lake.

Q.10 · Answer (B) Padma Shri
The Kuchipudi guru Vempati Chinna Satyam received the Padma Bhushan (1998), not the Padma Shri; the Sangeet Natak Akademi Award, the Kalidas Samman and an honorary doctorate were among his other honours.

Q.11 · Answer (B) Nitin Gadkari
Nitin Gadkari has been Union Minister of Road Transport and Highways since 2014.

Q.12 · Answer (C) Ravi Shankar
SSC's key gives Ravi Shankar, the sitar maestro, whose 1968 autobiography is 'My Music, My Life' (the question has the title the other way round).

Q.14 · Answer (D) Fugdi
Fugdi, danced by Goan women in circles or rows to singing and clapping, is traditionally performed during Ganesh Chaturthi.

Q.15 · Answer (C) octagonal
Sultan Ghari, built by Iltutmish in 1231-32 for his son Nasiruddin Mahmud, has an octagonal grave chamber below ground, whose roof forms an octagonal platform in the courtyard.

Q.16 · Answer (A) 12th March 2021
Azadi Ka Amrit Mahotsav began on 12 March 2021, the anniversary of the Dandi March, starting a 75-week countdown to the 75th anniversary of independence on 15 August 2022.

Q.17 · Answer (A) Promoting digestion and bowel regularity
Dietary fibre (roughage) adds bulk to food, helping digestion and the regular passage of stools.

Q.18 · Answer (C) Allowances and emoluments of the President are charged under it.
The President's emoluments and allowances are charged on the Consolidated Fund of India (Article 112), not the Contingency Fund. The Contingency Fund, set up under Article 267 for unforeseen expenditure, is held at the President's disposal.

Q.19 · Answer (A) single citizenship
Single citizenship is a unitary feature; the division of powers, bicameralism and a written constitution are federal ones.

Q.21 · Answer (B) Seoul, 1986
At the 1986 Asian Games in Seoul PT Usha won gold in the 200 m, 400 m, 400 m hurdles and 4 × 400 m relay, setting Games records in each, and silver in the 100 m.

Q.22 · Answer (A) Bengal
Fa-Hien sailed home from Tamralipti, the port in Bengal, by way of Sri Lanka.

Q.23 · Answer (A) Rajasthan
The loo is the hot, dry wind that blows over north and north-western India, Rajasthan among them, in May and June.

Q.24 · Answer (D) Athens 2004
Rajyavardhan Singh Rathore won silver in the men's double trap at Athens 2004, India's first Olympic medal in shooting.

Q.25 · Answer (A) Agra gharana
Lalith (Lalita) J. Rao was a Hindustani vocalist of the Agra gharana, a disciple of Ustad Khadim Hussain Khan.

English Language and Comprehension

Q.1 · Answer (A) Profligate
'Frugal' means sparing with money; its opposite is 'profligate', recklessly extravagant. 'Thrifty' is a synonym.

Q.2 · Answer (D) Alka bought new pair of shoes for the party.
'Pair' is a singular countable noun and needs an article: 'Alka bought a new pair of shoes for the party.'

Q.3 · Answer (A) Admited
The correct spelling is 'admitted': the 't' is doubled before '-ed'.

Q.4 · Answer (B) trained tirelessness
The verb 'trained' needs an adverb, 'tirelessly', not the noun 'tirelessness'.

Q.5 · Answer (D) leaving
A present participle introduces the result: 'caused extensive damage to homes, leaving many families homeless'.

Q.6 · Answer (D) Suspicious
The correct spelling is 'suspicious'.

Q.7 · Answer (D) lighting up the earth with
'Illuminating' means 'lighting up', so 'lighting up the earth with its gentle glow' keeps the meaning; the other options change it.

Q.8 · Answer (C) To bribe someone
To 'grease someone's palm' is to bribe them.

Q.9 · Answer (D) Esoteric
Besides 'airtight', 'hermetic' means obscure and understood only by a few: 'esoteric'.

Q.10 · Answer (C) for the music.
Music in general takes no article: 'her passion for music'.

Q.11 · Answer (C) Malodorous
'Ambrosial' means exceptionally pleasing in taste or smell; its opposite is 'malodorous', foul-smelling.

Q.12 · Answer (B) He way lying at
One lies 'in' or 'on' a bed, not 'at' it: 'He was lying in his bed'. (The sheet prints this option as 'He way lying at'.)

Q.13 · Answer (C) Dispersed
To 'garner' is to gather or collect; its opposite is 'dispersed', scattered.

Q.14 · Answer (B) Added fuel to the fire
To 'add fuel to the fire' is to make a bad situation worse.

Q.15 · Answer (D) an English literature
Names of languages and subjects take no article: 'an expert in English literature'.

Q.16 · Answer (C) Please
The correct spelling is 'please'.

Q.17 · Answer (D) put out
To 'put out' a fire is to extinguish it; to 'put up with' something is to tolerate it.

Q.18 · Answer (D) Resurrect
To 'resurrect' is to bring back to life.

Q.19 · Answer (C) feathers were as white as
The plural 'feathers' needs a plural verb, and the past 'was envious' calls for the past tense: 'her feathers were as white as'.

Q.20 · Answer (C) Bright
'Vivid' means intensely bright or clear: 'bright'. 'Dull' is its opposite.

Q.21 · Answer (A) are
The plural 'firsts' takes 'are', and the sentence states a general truth in the present: 'There are some firsts that stay in history forever.'

Q.22 · Answer (A) bigger
The 'Pocket Dynamo' was small, yet his deeds dwarfed those of 'far bigger' athletes; 'far' goes with a comparative.

Q.23 · Answer (D) priority
Funding sportspersons was not really a 'priority' for a country just finding its feet.

Q.24 · Answer (C) wrestled
Jadhav, a wrestler, 'wrestled' his way to fame: he fought hard for it.

Q.25 · Answer (D) claimed
He 'claimed', that is won, the bronze in the bantamweight category.

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